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SPACE SYSTEMS &

AEROSPACE CYBERSECURITY

UNDERSTANDING COMPONENTS, THREATS & SECURITY IN SATELLITE SYSTEMS

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KEY LEARNING OBJECTIVES

- What aerospace cybersecurity is and why it matters
- Space, ground and user segments, and how data flows
- Vulnerabilities and common attacks
- A real case: Black Sea, 2017
- How encryption, authentication and other controls reduce risk
- Standards, open-source tools and cloud architecture
- A signal simulation demo, and how systems recover

WHAT IS AEROSPACE CYBERSECURITY — AND WHY IT MATTERS

Aerospace cybersecurity means protecting satellites, ground stations and the communication links between them from cyber attacks.

It is not just internet security. It covers:

- Radio signals travelling from satellites to Earth
- Commands sent from ground stations up to satellites
- Software that controls the space systems

WHY IT MATTERS?

Because an attack on a satellite can affect many people who depend on satellite services

- Satellite services support navigation ,communication and precise timing.
- Attacks can mislead navigation of ships and aircraft
- Attacks can disrupt service that depend on satellite link or timing

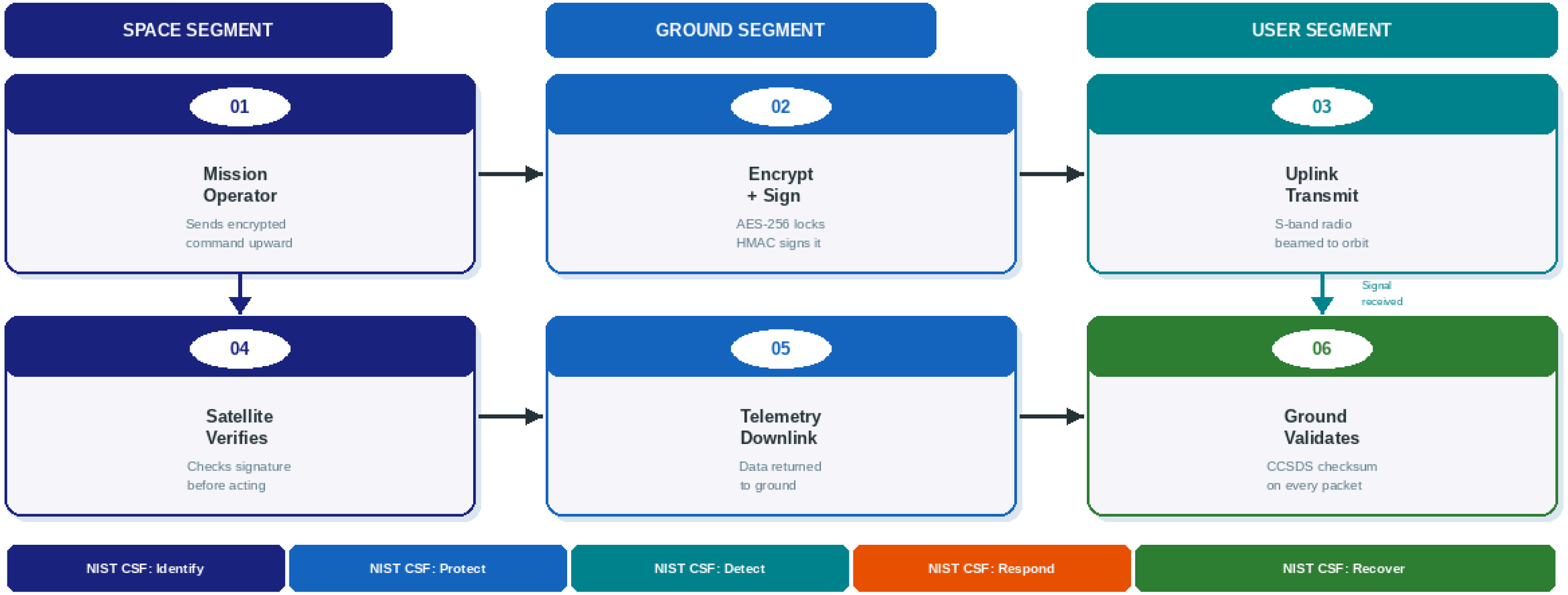
SATELLITE & AEROSPACE SYSTEM ARCHITECTURE

This simplified model has three segments. Each segment has different risks.

SPACE SEGMENT	GROUND SEGMENT	USER SEGMENT
• Satellites in orbit • Transponders— receive,amplify and retransmit signals • GNSS satellites broadcast precise timing signals **⚠ unauthorized commands, compromised software**	• Mission control centres and uplink stations talk directly to satelite • IT and network infrastructure • Main cyberattack — Ground systems are a major target **⚠ Target: Network hacking and command injection**	• Phones, aircraft, ships,GPS receivers • Legacy civil GPS has no built-in authentication • Some receivers can be spoofed **⚠ Target: GPS spoofing and signal interception**

HOW SATELLITE DATA FLOWS — AND HOW WE SECURE IT

This is a simplified example of a protected command and telemetry flow.



Security at Every Step:

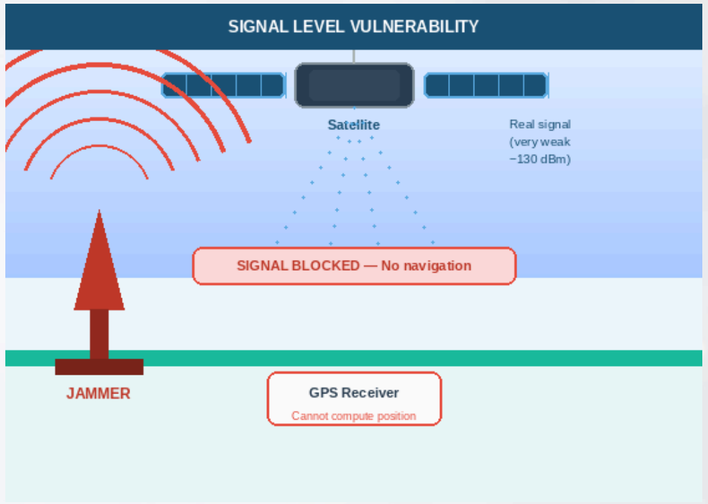
- AES-256 Encryption
- HMAC-SHA256 Signature
- mTLS Authentication
- CCSDS CRC Validation

VULNERABILITIES IN SPACE SYSTEMS

SIGNAL LEVEL

- Extremely weak satellite signals
- No built-in signal authentication

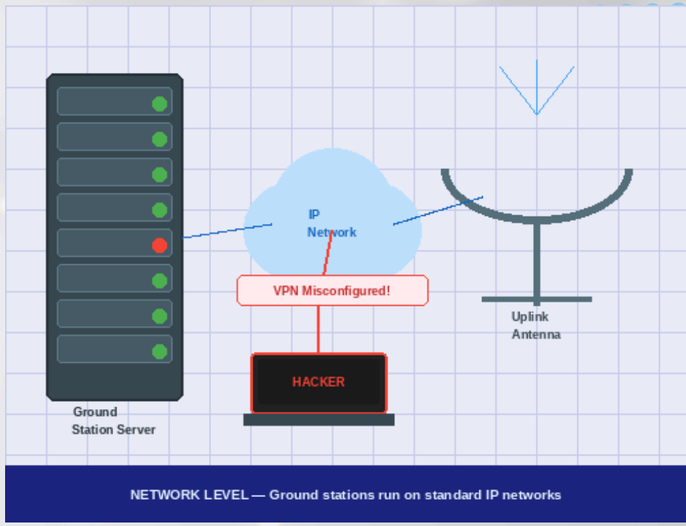
WHY dangerous:Low-cost transmitters can overpower weak GPS signals



NETWORK LEVEL

- Ground stations run on standard IP networks
- If the ground network is hacked ,attackers can disturb Satellite operations

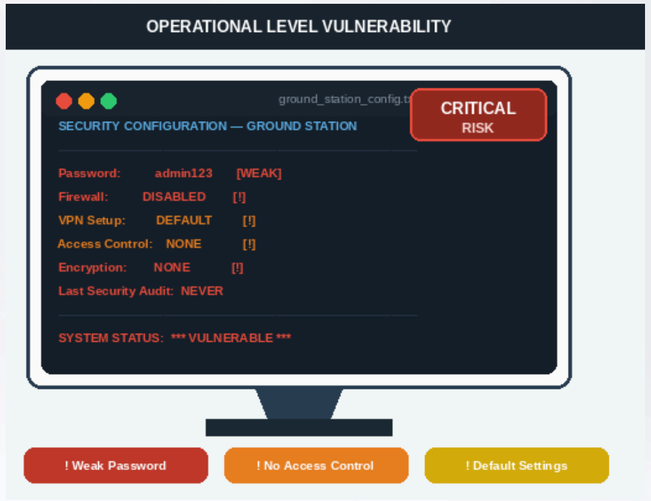
WHY dangerous: ViaSat 2022 — attackers entered via a misconfigured VPN device

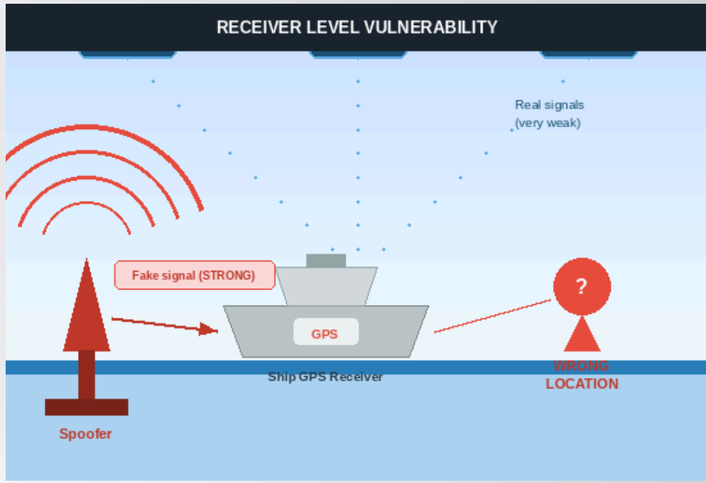


RECEIVER LEVEL

- Device lock to stongest signal
- Vulnerable firmware, poor validation

WHY dangerous: A vulnerable receiver may show misleading position or time without a clear warning.





OPERATIONAL LEVEL

- Weak keys and passwords
- Poor configuration of security settings

WHY dangerous: Misconfiguration can expose systems

COMMON CYBER ATTACKS ON SPACE SYSTEMS

1 **COMMAND INJECTION**	• Attacker sends a fake command to the satellite • This can disrupt or damage satellite operations •
2 **JAMMING**	• Strong noise blocks real GPS signal • Navigation can be disrupted
3 **SPOOFING**	• Fake signal stronger than real one • Receiver shows completely wrong locations
4 **SIGNAL HIJACKING**	• Signal intercepted before reaching satellite • Attackers take full control remotely

Signal Level Attack — How GPS Spoofing Works

Real signal weak

- Travels far ,low amplitude

Fake signal strong

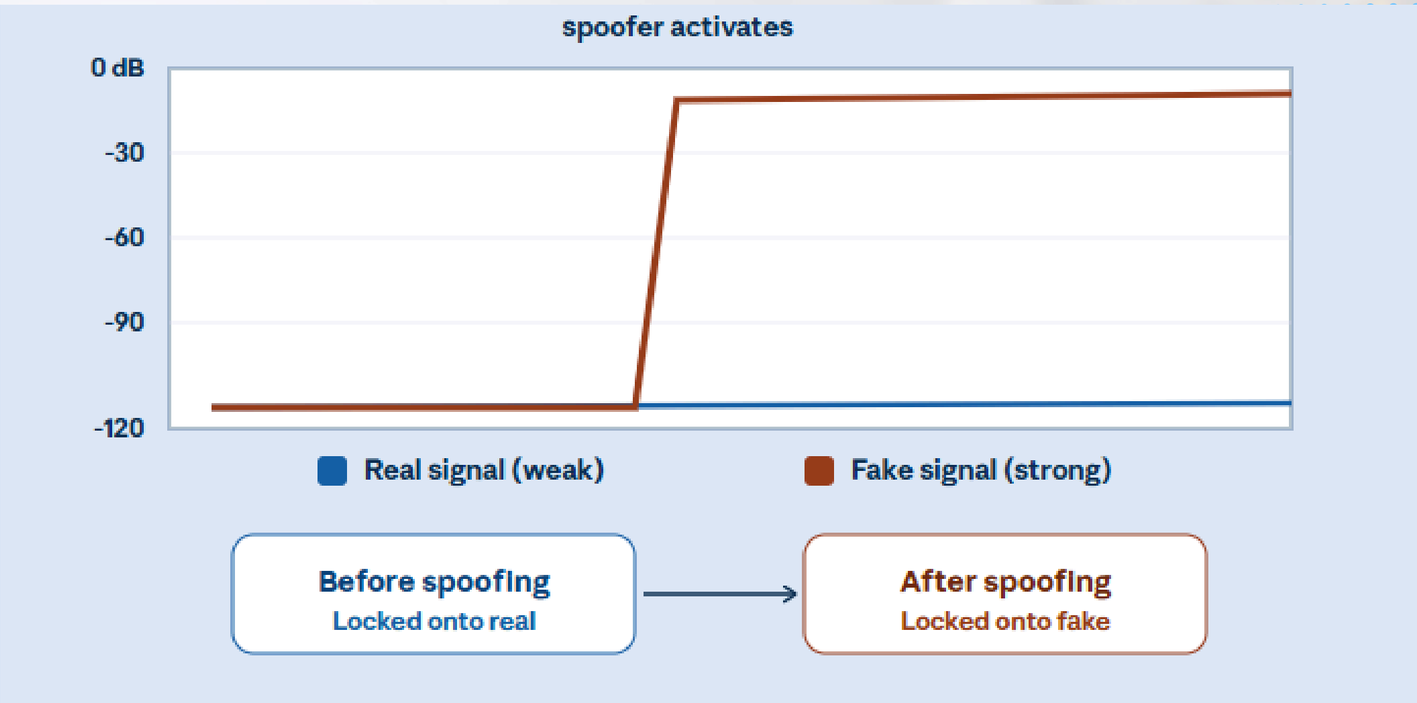
- Spoofers send bigger signal

Strongest signal always wins

- Receiver has no way to check

Wrong location shown

- Device shows wrong position, No warning



REAL-WORLD CASE STUDIES

BLACK SEA SPOOFING ATTACK (2017)

What Happened:

- 24 + ships showed wrong GPS location at Gelendzhik Airport
- Fake signal beat real satellite

Impact:

- Unreliable navigation information

Why it Matters:

- Gps Spoofing is real threat
- Space systems affect real world safety

Key Lesson:

- Use multiple satellite system and keep back up navigation ready
- cross-check with independent navigation

Reference:

https://insidegnss.com/reports-of-mass-gps-spoofing-attack-in-the-black-sea-strengthen-calls-for-pnt-backup/

ENCRYPTION AND AUTHENTICATION

ENCRYPTION

- Scrambles satellite commands
- AES-256 protects all

AUTHENTICATION

- Verifies every command source
- Unauthenticated commands should be rejected

WHY BOTH MATTER

- Encryption hides the message
- Authentication proves who sent it
- Encryption and authentication reduce risk, but cannot guarantee that a system is impossible to compromise.

AEROSPACE CYBERSECURITY FRAMEWORKS & STANDARDS

CCSDS

- Space communication standards
- Used by NASA and ESA

NIST SP 800-53

- Catalogue of security controls
- Controls access and audit logs

ISO/IEC 27001

- Manages full security programme
- Certifies agency security systems

ISO/IEC 27033

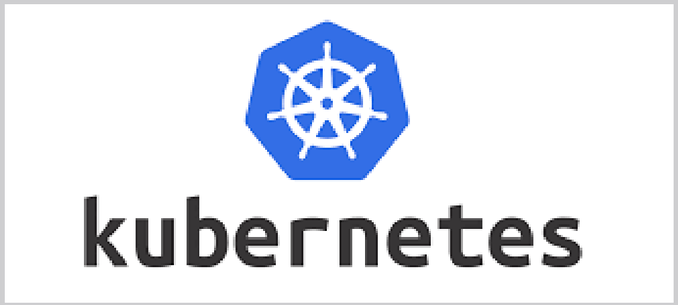
- Secures ground network links
- Guidance for network security.

ECSS-E-ST-80C

- Security in the space system life cycle.
- Risk assessment and security management.

OPEN SOURCE TOOLS

- Process satellite signals
- Simulate jamming attacks
- Show real time jamming attack



- Manages cloud applications
- Kubernetes access rules restrict access
- Unauthorized network access

- Global ground station network
- Monitors satellites worldwide
- Detects signal anomalies

Cloud Architecture in Space Systems

Data to cloud

- Ground station uploads satellite data

Cloud keeps safe

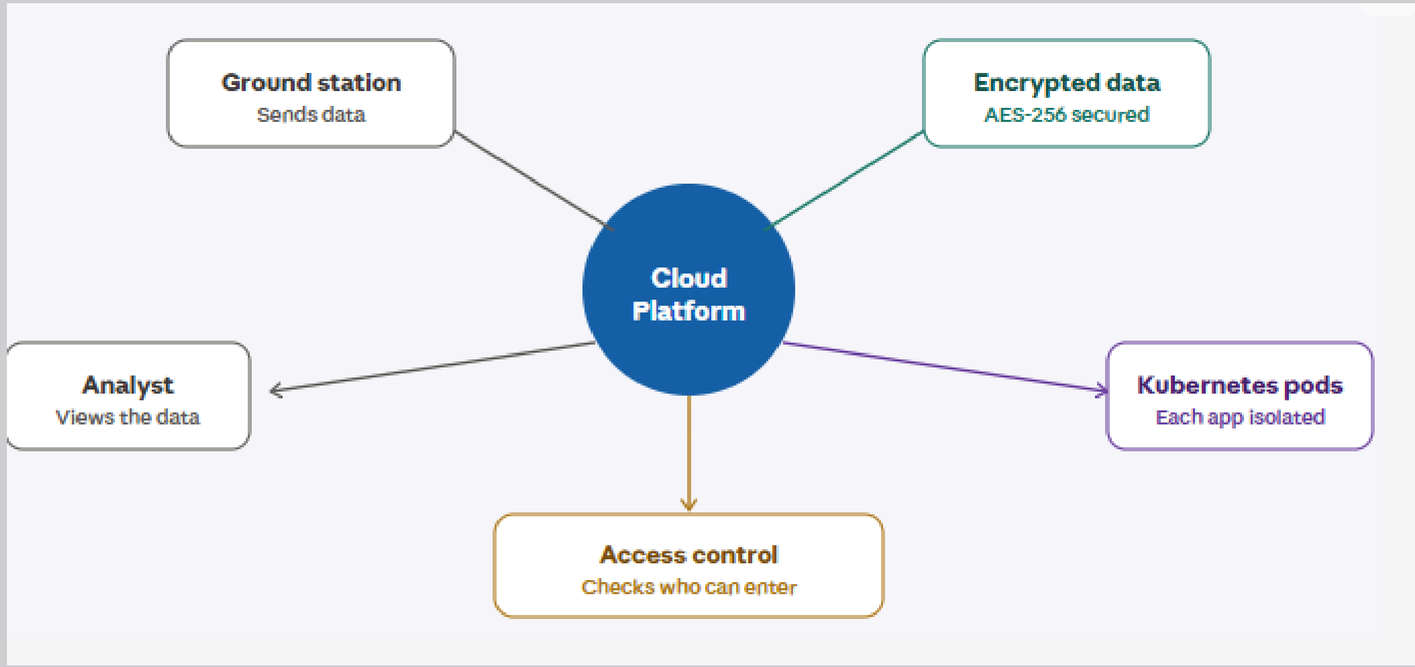
- Encryption protects data at rest

Apps stay separate

- Kubernetes isolates each one

Access control

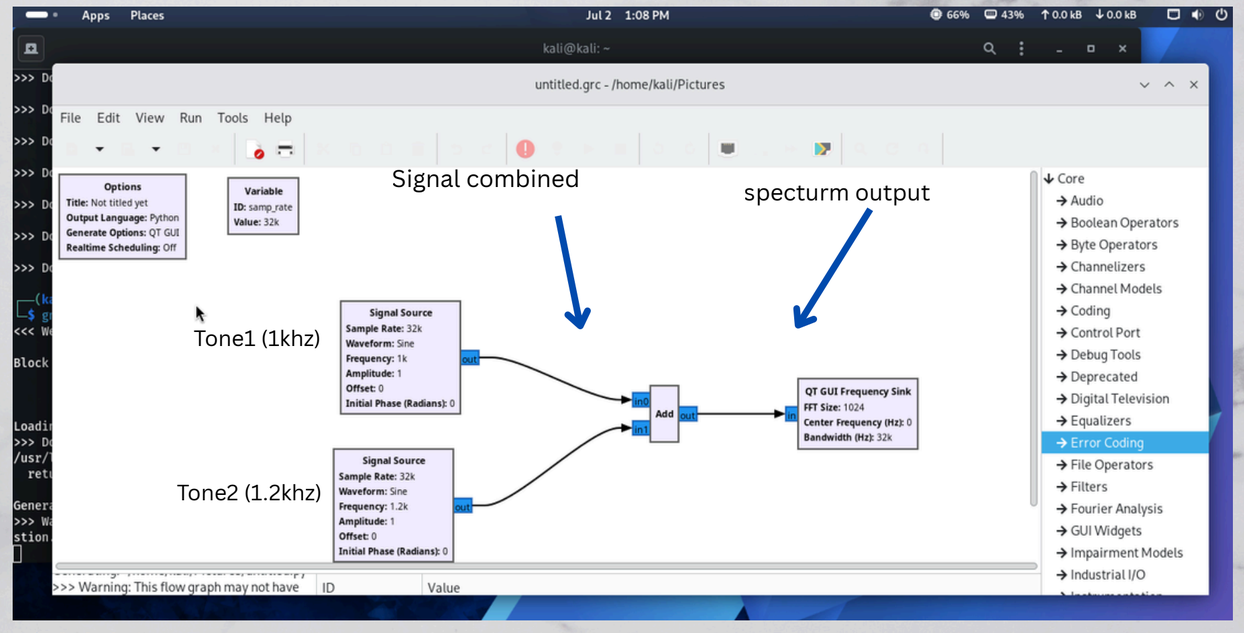
- only allowed users get in



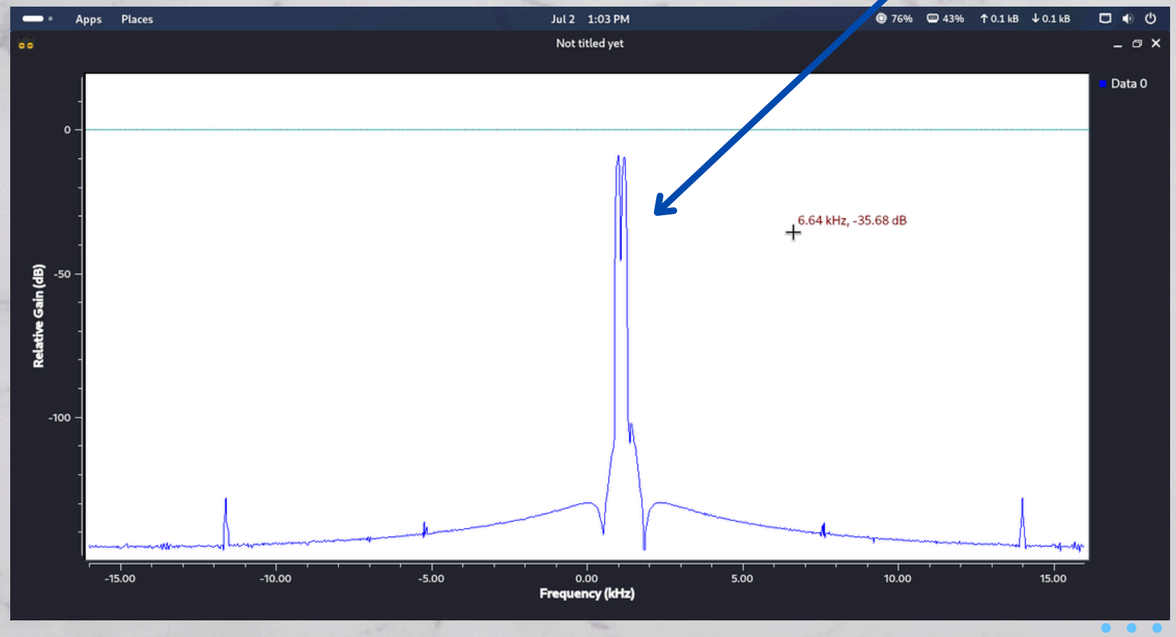
Satellite Signal - Spoofing Simulation

Simplified concept demo using GNU Radio on Kali Linux

Two-Tone Signal Spoofing Simulation



Spectrum Analysis Output



- Tone 1 (1 kHz) = genuine signal model. Tone 2 (1.2 kHz) = fake signal model
- A peak alone does not prove spoofing
- Tool used: GNU Radio Companion on Kali Linux
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CYBER ATTACKS — FAILURE & RECOVERY IN SPACE

What breaks down — and how space systems fight back

ATTACK TYPE	FAILURE — What Goes Wrong	RECOVERY — What We Do
SUPPLY CHAIN ATTACK *Malware planted before satellite launch*	**Malware activates in orbit** Satellite cannot be fixed remotely **Satellite permanently damaged**	**Firmware verified before launch** Vendor security strictly checked **Supply chain fully audited**
INSIDER THREAT *Trusted employee leaks system access*	**Attacker enters the system** Trusted access completely misused **System fully compromised**	**RBAC limits access rights** Every action logged and monitored **Zero trust policy enforced**
DENIAL OF SERVICE *Ground station flooded with fake requests*	**Operators lose satellite contact** Ground station completely overloaded **Aviation navigation disabled**	**Backup control centre activated** Traffic filtered immediately **Multi-frequency receiver used**
SIGNAL INTERFERENCE *Satellite data corrupted during transmission*	**Wrong readings sent to ground** Satellite data fully corrupted **Vessel drifts off course**	**CCSDS checksum detects error** Corrupted packets flagged instantly **Signal anomalies monitored**

Defence-in-Depth: Signal Layer → Network Layer → Receiver Layer → Operational Layer | NIST CSF: Protect · Detect · Respond

Defending space -our Countermeasures

Secure Ground Network

- Keep control network separate from office
- Segmentation reduces the chance of spread
- Restrict access to authorized users

Protect Satellite Signals

- Encryption (e.g., AES) protects command content
- Unauthorized commands should be rejected automatically
- HMAC authentication tag verifies every command

Supply Chain Security

- All hardware checked before launch
- Reduces risk of untrusted software
- Make sure firmware is safe and clean
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CONCLUSION

Space systems are part of daily life. GPS, banking, and flights all depend on satellites working safely.

Every part of the system can be attacked — space, ground, or the user side. That's why encryption alone is not enough. We also need authentication, so unauthorized commands can be rejected.

These are not just theory. Black Sea already showed us this risk is real.

Frameworks like NIST, CCSDS, ISO, and ECSS give us a standard to follow. And recovery plans matter just as much as prevention, because attacks can still happen even with good security.

THANK YOU